

Pavit Thakur

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Education

Texas A&M University – BS in Electrical Engineering (3.81) – Dean's Honor Roll

May 2028

Experience

Avionics Engineer, Texas A&M Sounding Rocketry Team

Incoming Aug 2026

- Selected to engineer full-stack avionics solutions, with upcoming responsibilities including custom PCB design and embedded systems programming for real-time flight telemetry.

Power Distribution Engineer, Texas A&M Solar Car Racing

Aug 2025 – Present

- Created a synchronous 12-5V buck converter in Altium designer to use a high-voltage power supply with low-voltage communication systems.
- Established a CAN-BUS communication protocol on a MITSUBA motor controller to retrieve the RPM, motor current, and other telemetry data from the motor.
- Designed and soldered the 2-layer Distributed Control Systems (DCS) PCB which connected CAN communication lines, throttle, the SPI module, cruise control, lighting, and various other control IO.

Hardware Engineer, IEEE x Aggie Data Science Club EMI Detection Project

Feb 2026 – April 2026

- Implemented AC coupling, level shifting, a low noise amplifier, a low pass filter, and an ADC to build a probing array and signal conditioning system to detect EM interference from electronics with a nine-person team.

FPGA Design Intern, MIT Beaver Works Summer Institute

May 2024 – Aug 2024

- Collaborated in a 5-person team to architect a custom ASIC and PCB for energy-efficient LED matrix control, utilizing Verilog HDL for RTL design and logic synthesis.
- Optimized control overhead to achieve a 30% power reduction through custom Verilog-based SPI design.
- Executed a complete RTL-to-GDSII flow using OpenLane, including 3D layout inspection/Verilog logic synthesis.

Projects

Level 2 Solar-Powered EV Charger

- Engineered a solar-powered Tesla EV charger in a three-person team, integrating eight 240W solar panels, a 48V battery system, an AC/DC inverter, and a 2500W charge controller to deliver 2.4 kWh of energy.
- Coordinated local sponsor outreach, resulting in a \$4,000 technical supply donation.

Custom FPV Drone

- Designed and fabricated a custom FPV drone from scratch using open-source components, integrating a live-feed video transmitter and radio for remote communication.
- Utilized Fusion360 and Solidworks to engineer 3D-printed landing stilts, propeller guards, and a custom claw, earning 1st place at the TSA Texas State competition.

SkateCam

- Created a multiplayer skating game utilizing Computer Vision to process physical hand inputs via OpenCV and MediaPipe; mapped nodes on users' hands to enable real-time, head-to-head gesture control.
- Received 1st prize at Texas A&M's 2025 HowdyHack Hackathon.

Iron Man Helmet

- 3D modeled/printed 1:1 scale of Iron Man helmet using Fusion 360/Ultimaker Cura.
- Soldered, wired, and coded an Arduino nano to motorize the helmet; added working LED lights to eyes.

Skills

Python, Verilog HDL, OpenLane, Altium Designer, C++ , Docker, SolidWorks, Fusion 360, Autodesk Inventor, Ultimaker Cura, Betaflight